



Transfer Learning with Particle Swarm Optimization for Durian Leaf Disease Image Classification

Tran Nguyen Phi Hung¹ · Nguyen Minh Tuan¹

Received: 17 December 2025 / Accepted: 9 April 2026

© The author(s), exclusively licensed to Springer-Verlag GmbH Germany, a part of Springer Nature 2026

Abstract

Timely durian leaf disease detection is critical for Vietnam's agricultural productivity, yet traditional methods remain labor-intensive and error-prone. This study proposes a hybrid pipeline integrating deep transfer learning with binary Particle Swarm Optimization (PSO) for efficient disease classification. Three lightweight backbones, such as MobileNetV3-Large, EfficientNet-B0, and EfficientNetV2-B0 to extract 128-dimensional features from a real-world Vietnamese durian dataset (2595 images, 6 classes), which PSO then prunes using five-fold support vector machine (SVM) cross-validation fitness. The optimized subsets were evaluated across five machine learning (ML) classifiers, achieving up to 92.6% test accuracy with a modest improvement over the baseline while reducing dimensionality. PSO-selected features demonstrated the potential for accelerated inference and interpretable agricultural diagnostics on resource-constrained devices.

Keywords Particle Swarm Optimization · Durian Leaf Disease · Image Classification · MobileNet · EfficientNet

Introduction

Durian (*Durio zibethinus*) is widely recognized as one of the most significant fruit crops in Southeast Asia and is often referred to as the “king of fruits” due to its distinctive qualities and economic importance. The cultivation is prominent in countries such as Thailand, Malaysia, Indonesia, Vietnam, and the Philippines, with global demand steadily rising over the past decade. Beyond the commercial value, durian fruit is a rich source of carbohydrates, polyphenols, flavonoids, and bioactive compounds such as glutathione and glutamylcysteine, offering notable antioxidant capacity and health benefits for consumers (Pinsorn and Sirikantaramas 2025).

Traditional approaches for disease identification, such as visual assessment, laboratory analysis, or advice from experts, come with substantial challenges. Such manual proce-

dures are typically slow, costly, and may be inaccurate if the evaluator lacks the necessary training. In some cases, misinterpretation of symptoms results in the misuse of pesticides, creating risks for people and the environment. (Das et al. 2025). Machine learning (ML) and deep learning have revolutionized smart agriculture, offering fast and accurate solutions for plant disease diagnosis. Deep convolutional neural networks (CNNs) excel at analyzing leaf images, automatically identifying disease patterns with remarkable precision that often surpasses human performance. These advances make it feasible to develop affordable, user-friendly diagnostic tools, empowering farmers to detect plant diseases independently without expert intervention.

This study implements Particle Swarm Optimization (PSO) on deep learning feature subsets with a focus on practical application to a real-world durian leaf image dataset collected from orchards in Binh Phuoc and Cai Lay provinces, Vietnam. The research is guided by two primary objectives. First, we survey and evaluate multiple lightweight and advanced deep learning backbones (MobileNetV3-Large, EfficientNet-B0, EfficientNetV2-B0) on a real-world Vietnamese durian leaf disease dataset. Second, we experiment with binary PSO with support vector machine (SVM) fitness to select optimal feature subsets from these backbones, then feed them into various ML classifiers for comparison against standard end-to-end deep

✉ Nguyen Minh Tuan
minhtuan@ptit.edu.vn

Tran Nguyen Phi Hung
n21dccn036@student.ptithcm.edu.vn

¹ Department of Information Security, Faculty of Information Technology, Posts and Telecommunications Institute of Technology, 11 Nguyen Dinh Chieu, 700000 Sai Gon ward, Ho Chi Minh City, Viet Nam

learning baselines across accuracy, precision, recall, and F1-score metrics (Thanh et al. 2025).

Related Work

Early research on automated plant disease detection primarily utilized classical ML algorithms, which required manual feature extraction and preprocessing. Techniques such as SVM, k-nearest neighbors (KNN), decision trees, and shallow artificial neural networks were widely adopted. These methods relied heavily on handcrafted features based on color, texture, or shape, and often performed well in controlled environments with small datasets. However, their effectiveness was limited by the need for careful feature engineering and their sensitivity to variations in real-world conditions, motivating the development of more advanced methods in subsequent years (Bhagwat and Dandawate 2021).

Deep learning methods, especially CNNs, have significantly advanced plant disease detection. For example, Baser et al. (2023) developed TomConv, a lightweight CNN architecture built from scratch to diagnose diseases in tomato leaves. The model employs four convolutional layers and demonstrates superior performance compared to several established CNN models, achieving a validation accuracy of 98.19%. Meanwhile, Paymode and Malode (2022) applied transfer learning with the VGG16 architecture, fine-tuning pre-trained weights on a multi-crop leaf disease dataset. Their approach resulted in high classification accuracy for both grape and tomato leaves, reporting 98.40% and 95.71%, respectively. These studies highlight that custom-designed and transfer learning-based CNN models, when tailored to the application, can both yield highly effective results in agricultural image classification.

Traditional CNN architectures built from scratch often demand extensive storage and computational resources, making them challenging to use on mobile or embedded platforms (Agarwal et al. 2020). This has created a strong interest in compact deep learning models, such as MobileNet, for real-time plant disease detection on smartphones. Introduced by Google in 2017, MobileNet is lightweight and efficient, designed for devices with limited resources. By employing depthwise separable convolutions, MobileNet reduces computation cost by about eight to nine times compared to standard convolutions, and significantly decreases the number of model parameters, while still delivering robust classification performance (Howard et al. 2017).

Practical applications of MobileNet have demonstrated the effectiveness in agricultural settings. Elhassouny and Smarandache (2019) leveraged MobileNet's depthwise separable convolutions to develop a real-time mobile applica-

tion for tomato leaf disease recognition, achieving 88.4% accuracy on standard smartphones. Subsequent evolution of the architecture led to MobileNetV3-small, which further improved performance by incorporating Squeeze-and-Excitation modules and optimized activation functions. Khan et al. (2023) demonstrated that MobileNetV3-small reached 99.50% accuracy on plant disease classification while employing dynamic quantization to reduce model parameters to 0.9 million, enabling efficient deployment on resource-constrained edge devices with minimal inference latency.

EfficientNet, introduced by Tan and Le in 2019 (2020), another lightweight architecture, revolutionizes CNN scaling through compound coefficients that uniformly balance network depth, width, and resolution, yielding state-of-the-art accuracy with far fewer parameters than predecessors such as ResNet. Rukuna et al. (2025) applied EfficientNet-B5 to citrus leaf diseases on a Kaggle dataset, pushing validation accuracy to 99.22% and test accuracy to 99.23% using SMOTE balancing and augmentation.

In summary, the evolution from traditional ML to deep learning (Tuan et al. 2023, Tuan et al. 2024, Tuan and Meesad 2025) and the shift toward lightweight architectures such as MobileNet and EfficientNet have enabled the development of practical plant disease diagnostic systems deployable on mobile devices, thereby offering farmers convenient, real-time disease management capabilities. However, most existing research relies on controlled laboratory datasets rather than real-world data collected from actual farm environments, which limits their practical applicability across diverse agricultural conditions.

Related Definitions

Particle Swarm Optimization

PSO is a population-based metaheuristic algorithm inspired by the collective social behavior observed in nature, such as bird flocking and fish schooling. Developed by Eberhart and Kennedy in 1995, PSO has become one of the most widely adopted swarm intelligence algorithms due to its algorithmic simplicity, computational efficiency, and robust performance across diverse continuous optimization problems.

The fundamental principle of PSO involves maintaining a population of candidate solutions, referred to as particles, which navigate through an n -dimensional search space following simple mathematical rules that govern their velocities and positions, as illustrated in Fig. 1. Each particle i represents a potential solution characterized by the current position $\mathbf{x}_i^t$ and velocity $\mathbf{v}_i^t$ at iteration t . The algorithm explores both individual experience through personal best

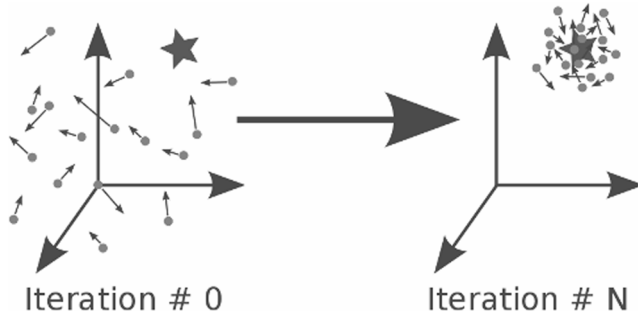


Fig. 1 Particle Swarm Optimization visualization (Pagmo 2024)

positions $\mathbf{p}_i$ and collective intelligence via the global best position $\mathbf{g}$ discovered by the entire swarm.

The PSO algorithm iteratively updates particle states through three core mechanisms: inertia-based momentum, cognitive learning from personal experience, and social learning from swarm knowledge. At each iteration t , the velocity of particle i is updated according to:

$$\mathbf{v}_i^{t+1} = \omega \mathbf{v}_i^t + c_1 r_1 (\mathbf{p}_i^t - \mathbf{x}_i^t) + c_2 r_2 (\mathbf{g}^t - \mathbf{x}_i^t), \quad (1)$$

where ω denotes the inertia weight controlling the influence of previous velocity and balancing exploration versus exploitation; c_1 and c_2 represent cognitive and social acceleration coefficients, respectively, determining the relative importance of personal and global best positions; and $r_1, r_2 \in [0, 1]$ are uniform random variables introducing stochastic exploration to prevent premature convergence.

Subsequently, the particle position is updated using the newly computed velocity:

$$\mathbf{x}_i^{t+1} = \mathbf{x}_i^t + \mathbf{v}_i^{t+1}. \quad (2)$$

After position updates, each particle evaluates the objective function $f(\mathbf{x}_i^{t+1})$ and updates the personal best position if improvement is observed:

$$\mathbf{p}_i^{t+1} = \begin{cases} \mathbf{x}_i^{t+1}, & \text{if } f(\mathbf{x}_i^{t+1}) < f(\mathbf{p}_i^t), \\ \mathbf{p}_i^t, & \text{otherwise.} \end{cases} \quad (3)$$

The global best position is then determined as the best personal position across all particles:

$$\mathbf{g}^{t+1} = \arg \min_{\mathbf{p}_i^{t+1}} f(\mathbf{p}_i^{t+1}). \quad (4)$$

The complete PSO procedure can be summarized as follows:

1. Initialize particle positions $\mathbf{x}_i^0$ and velocities $\mathbf{v}_i^0$ randomly within the search space.
2. For each iteration t until a termination criterion is met:
 - (i) Update velocities according to (1).
 - (ii) Update positions according to (2).
 - (iii) Evaluate objective function $f(\mathbf{x}_i^{t+1})$ for each particle.
 - (iv) Update personal best positions using (3).
 - (v) Update global best position using (4).
3. Return the global best position $\mathbf{g}$ as the optimal solution.

Methods

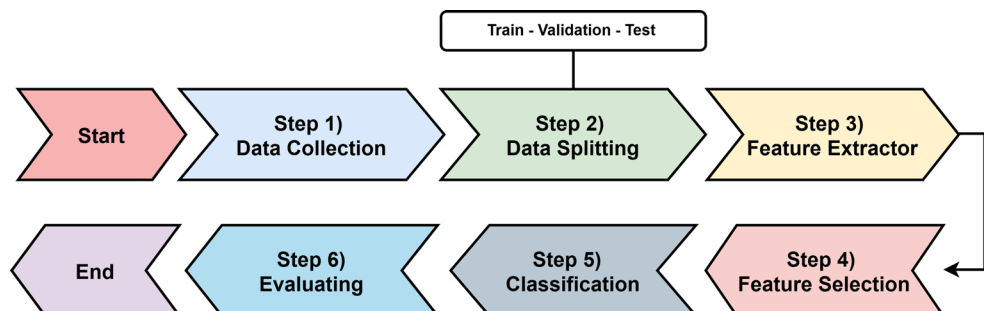
Workflow

Figure 2 illustrates the complete pipeline for durian leaf disease classification through a systematic six-stage process. The methodology follows a sequential execution path from initialization to completion, processing a multi-class dataset comprising six distinct durian leaf categories. The experimental framework encompasses the following procedural stages:

Data Collection Collect durian leaf images from Vietnamese orchards; the dataset contains 2595 high-resolution images across six classes: Healthy, Algal, *Colletotrichum*, *Phomopsis*, *Rhizoctonia*, Blight.

Data Partitioning The dataset is pre-split into train (70%), validation (15%), and test (15%) folders, ensuring robust model training, tuning, and unbiased evaluation.

Fig. 2 Implementation process flowchart



Feature Extraction Pre-trained MobileNetV3-Large (also EfficientNetB0/V2-B0) with transfer learning: resize $400 \times 400 \rightarrow 224 \times 224$, remove top layer, add custom 128-unit dense+dropout+softmax classification head.

Feature Selection Extract 128-dimensional features from the final dense layer (train+validation merged); apply binary PSO+SVM (five-fold CV multi-objective fitness with sparsity penalty) to select optimal subset.

Classification Selected features are fed to lightweight ML classifiers (Logistic Regression, SVM-RBF, XGBoost, Random Forest, Voting Ensemble) and rigorously validated on a held-out test set.

Evaluating The pipeline above is then compared to the baseline transfer learning model formulated in the “Feature Selection” step, in terms of accuracy, precision, recall, and F1-score to assess the performance.

Dataset

The durian leaf dataset used in this study is based on the published Mendeley dataset (Thanh et al. 2025). This dataset represents a comprehensive collection of durian leaf images specifically designed for agricultural diagnosis and automated plant disease detection in Vietnam. It contains 2595 high-quality images of durian leaves across six classes: Healthy, Algal, *Colletotrichum*, *Phomopsis*, *Rhizoctonia*, and Blight. Each image has a resolution of 400×400 pixels (Fig. 3).

Before training, several preparation steps are performed to ensure compatibility with MobileNetV3 and effective model training:

- **Image Resizing:** All images are resized to 224×224 pixels, matching the input size expected by MobileNetV3. This size balances detail preservation and computational efficiency.
- **Dataset Splitting:** The dataset is already divided into 70% for training and 15% for validation and 15% for testing.
- **Batch Processing:** The images are grouped into batches of 32 to optimize memory and computation during training (Table 1).

Feature Extraction

The custom architecture is constructed as a sequential model, as depicted in Fig. 4. Initially, we built the full model end-to-end (normal transfer learning), including the MobileNetV3 backbone and a custom Classification Head, to compare the performance with our pipeline strategy (Feature Extraction $\rightarrow$ Feature Selection $\rightarrow$ Classification). After training, we extract the final dense layer (128 units) for feature selection. This approach allows us to leverage the full model’s feature extraction capabilities while optimizing the pipeline for specific tasks.

1. **Feature Extraction Backbone:** We selected MobileNetV3 (then EfficientNetB0 and EfficientNetV2-B0) as our base model due to the high efficiency and proven performance

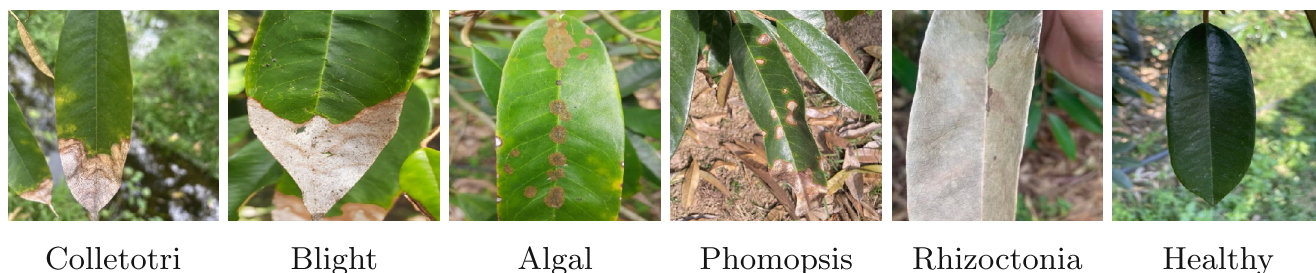


Fig. 3 Samples from the Vietnam durian dataset

Table 1 Durian leaf disease dataset distribution. This table shows the distribution of all images across the six classes that the model will use for training, validation, and testing

Class name	Number of images	Training (70%)	Validation (15%)	Test (15%)
Healthy	484	338	72	74
Algal	462	323	69	70
<i>Colletotrichum</i>	400	280	60	60
<i>Phomopsis</i>	411	287	61	63
<i>Rhizoctonia</i>	398	278	59	61
Blight	440	308	66	66
Total	2595	1814	387	394

Fig. 4 The architecture of the custom model. The image uses MobileNetV3 for main visualization; other backbone models (EfficientNet-B0, EfficientNetV2-B0) also adopted the same methodology. *Global-Avg2D* Global Average Pooling 2D, *FC* Fully Connected layer, *NL* Non-Linearity, *Dwise* Depth-wise Convolution, *ReLU* Rectified Linear Unit, *hard-s* Hard-Swish

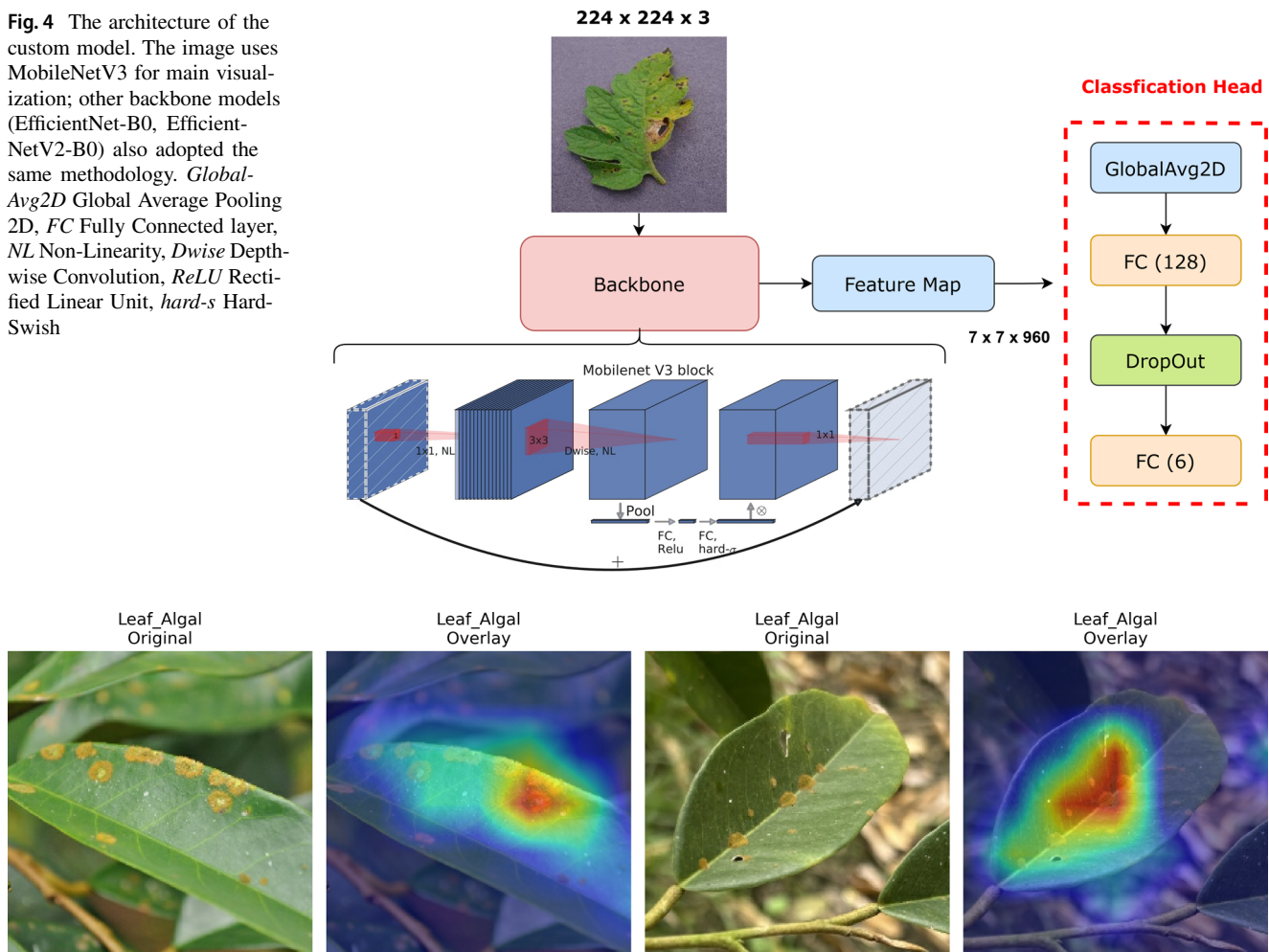


Fig. 5 Gradient-weighted Class Activation Mapping (Grad-CAM) visualization on disease regions. After the training process, the model generates predictions for leaf images. The Grad-CAM heatmap illustrates which regions the model considers most indicative of disease during the inference

learning to preserve generic visual features learned from ImageNet, while customizing the final layers to better adapt to our specific task. This strategy retains the benefit of feature extraction while allowing targeted adaptation with a small learning rate.

2. **Classification Head:** A GlobalAveragePooling2D layer aggregates spatial information into a compact vector, which is passed to a Dense layer with 128 units (ReLU) for non-linear fusion, followed by a Dropout layer (rate 0.3–0.5) for regularization. A final softmax activation produces class probabilities over the six categories, providing a lightweight yet effective classifier compatible with mobile/edge constraints (Fig. 5).

Feature Selection

Given a feature set $F = \{f_1, f_2, \dots, f_n\}$ with n extracted features, the objective of PSO is to find the optimal subset $S \subseteq F$ that maximizes classification accuracy. We apply

binary PSO to solve this problem. Each particle i in the swarm is represented by a binary vector:

$$X_i = (x_{i1}, x_{i2}, \dots, x_{in}),$$

where $x_{ij} \in \{0, 1\}$ indicates whether feature f_j is selected ($x_{ij} = 1$) or not ($x_{ij} = 0$). The search space has size 2^n , where n is the total number of features. For example, if we have five features and a i particle represented as $X_i = (1, 0, 1, 0, 1)$, it means that features f_1, f_3 , and f_5 are selected while f_2 and f_4 are not, and the search space contains $2^5 = 32$ possible feature subsets we can explore (Fig. 6).

The velocity of particle iii at iteration $t + 1$ is updated using the standard Particle Swarm Optimization (PSO) velocity update equation described in the Basic Theory section.

$$V_i^{t+1} = wV_i^t + c_1 \cdot r_1 \cdot (pbest_i^t - X_i^t) + c_2 \cdot r_2 \cdot (gbest^t - X_i^t).$$

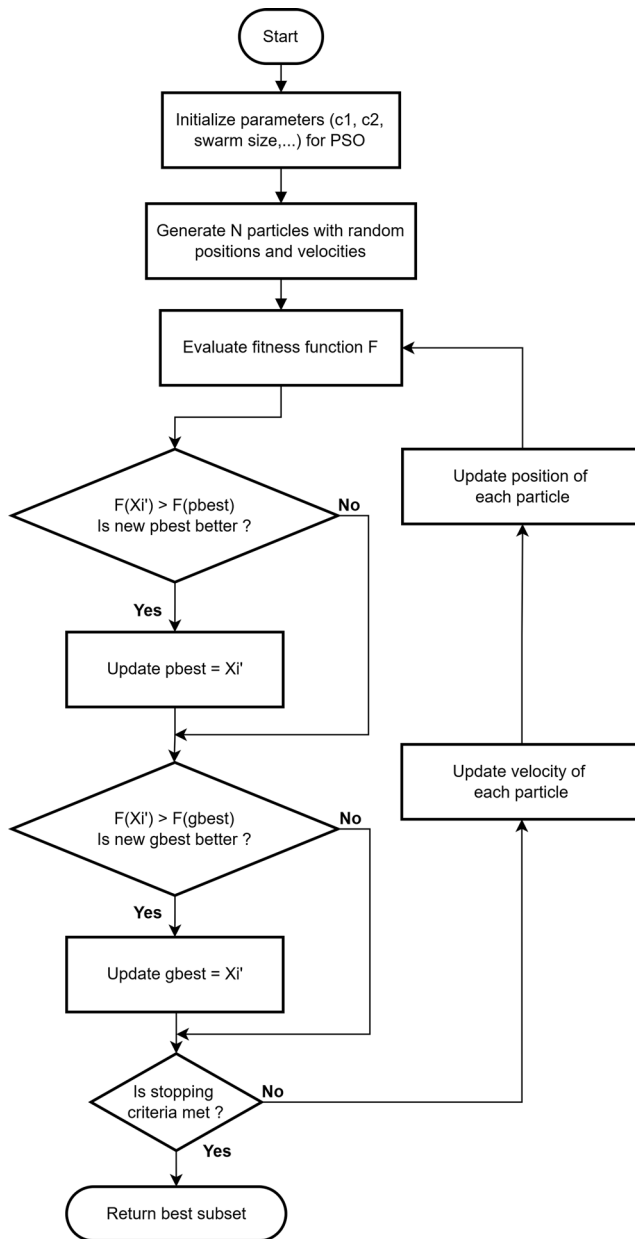


Fig. 6 Flowchart of feature selection process with Particle Swarm Optimization

To convert continuous velocity to binary position in PSO, we use the sigmoid transfer function:

$$T(v_{ij}^{t+1}) = \frac{1}{1 + e^{-v_{ij}^{t+1}}}$$

The new position is now determined by:

$$x_{ij}^{t+1} = \begin{cases} 1 & \text{if rand() < } T(v_{ij}^{t+1}), \\ 0 & \text{otherwise.} \end{cases}$$

The fitness of each particle is evaluated based on a multi-objective function that maximizes five-fold cross-validation accuracy (using a Support Vector Machine with Radial Basis Function [SVM-RBF] classifier) while minimizing feature count via a penalty term (0.05 multiplied by the feature ratio). To maximize data utilization, the original training and validation sets were merged for this stage to perform cross-validation. Particle velocities were clipped to a range of $[-6, 6]$ to prevent divergence. This function aims to maximize accuracy while minimizing feature count:

$$\max_{X_i} F(X_i) \text{ subject to } X_i \in \{0, 1\}^n,$$

where the optimal solution X^* represents the best feature subset that achieves the highest classification accuracy. The setup parameters are:

- Population size: $N = 60$
- Acceleration coefficients: $c_1 = c_2 = 0.75$
- Inertia weight: $w = 0.5$
- Maximum velocity: $V_{max} = 6.0$

Classification

The final stage of the pipeline aims to validate the robustness and generalization capability of the optimized feature set derived from the PSO stage. To ensure rigorous evaluation, we employed five-fold cross-validation on this optimal feature subset to compare the learning efficiency and stability across different classifiers. The evaluated models include XGBoost, SVM, Random Forest, Logistic Regression, and a hard-voting ensemble of these four classifiers using standard hyperparameter configurations. Following the cross-validation phase, the models were retrained on the complete training dataset and ultimately evaluated on the independent test set. This procedure effectively isolates model selection from final performance assessment, providing a reliable measure of the pipeline's efficacy.

Experiments

The environment used in this paper for the experimental setup was based on the Kaggle environment, with computational support from an NVIDIA T4 GPU. For the training phase, the model was configured with a batch size of 32, an input image size of 224×224 pixels, and was trained for a total of 30 epochs. In addition, we conducted the experiments on five different seeds (42, 64, 128, 256, and 512) to ensure robust comparison. The most used metrics include accuracy, precision, recall, and F1-score. Each provides a different perspective on classification performance,

especially in multi-class datasets (*the best seed on the test set will be chosen for comparison*).

Numerical Results

Feature Extraction—Phase 1

We report test accuracy and cross-entropy loss computed with Keras evaluate and provide per-class precision, recall, and F1-score along with macro and weighted averages using the scikit-learn classification report. The two sources are consistent; minor differences arise from aggregation and rounding.

During transfer learning, MobileNetV3-Large excelled in stability (train-validation gap 0.6%) for robust generalization, while EfficientNetV2-B0 showcased superior ca-

Table 2 Performance comparison of models (phase 1) on the test set. EfficientNet-B2 typically has higher capacity, learns richer features, and performs better after fine-tuning or optimization

Model (seed)	Accuracy	Macro F1
MobileNetV3-Large (512)	0.914	0.911
EfficientNet-B0 (42)	0.896	0.895
EfficientNetV2-B0 (42)	0.921	0.920

capacity (93.7% train accuracy, 0.22 loss) far outpacing others despite a 3.5% gap signaling mild overfitting. EfficientNet-B0 offered balanced middling performance. All three demonstrated strong overall learning capabilities (Fig. 7).

In Table 2, the test results reveal EfficientNetV2-B0 leading with 92.1% accuracy and 92.0% F1-score, showcasing superior feature extraction over the balanced EfficientNet-

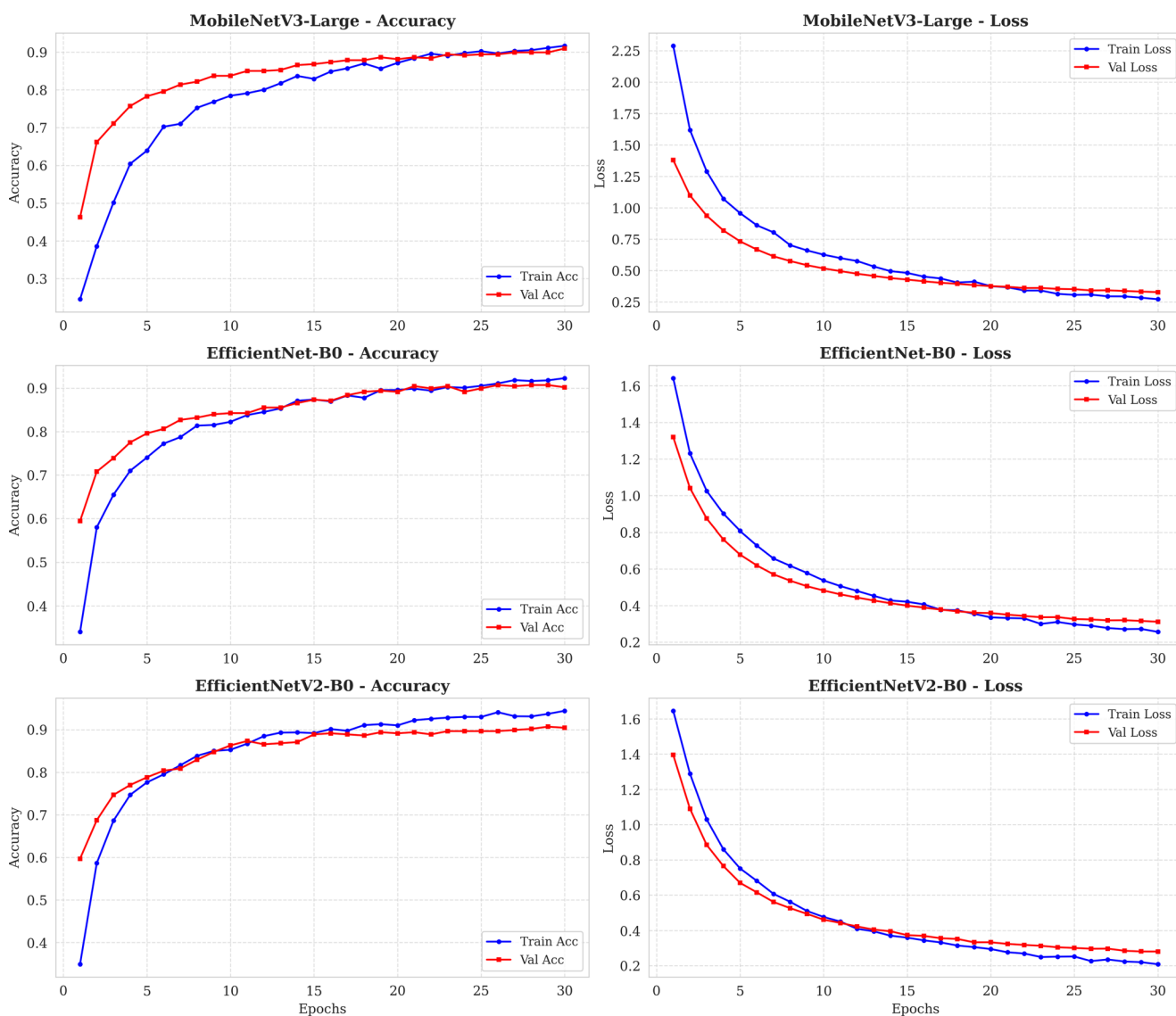


Fig. 7 Training and validation curves on MobileNetV3-Large, EfficientNetB0, and EfficientNetV2-B0—phase 1

Table 3 Particle Swarm Optimization feature selection results (phase 2)—best seed per model. This behavior arises because redundant feature representations in expressive models allow aggressive pruning without information loss, which simultaneously improves generalization and simplifies the optimization search space, leading to faster convergence

Model (seed)	Time (min)	CV fitness	Features	Reduction (%)
MobileNetV3-Large (512)	19.40	0.931	66/128	48.4
EfficientNet-B0 (42)	18.15	0.936	51/128	60.2
EfficientNetV2-B0 (42)	14.80	0.942	51/128	60.2

CV Cross-validation

B0 (89.6%) and stable MobileNetV3-Large (91.4%), providing the strongest baseline for PSO feature selection.

In detail, per-class analysis shows that Leaf_Healthy, Leaf_Rhizoctonia, and Leaf_Phomopsis demonstrated robust performance, while Leaf_Blight and Leaf_Colletotrichum exhibited significant challenges (lower recall in all models) with frequent mutual misclassification. This confusion pair highlights the need for PSO feature selection to extract discriminative patterns for improved classification (Fig. 8).

Feature Selection—Phase 2

PSO's ability to compress feature spaces by 48–60% while maintaining accuracy above 93% is shown in Table 3. EfficientNetV2-B0 exhibited superior optimization convergence with 94.24% fitness in just 14.8 min, outpacing both

MobileNetV3-Large (93.06%, 19.4 min) and EfficientNet-B0 (93.60%, 18.2 min). The inverse correlation between feature retention and fitness suggests redundancy in deeper architectures benefits more from aggressive pruning.

Classification—Phase 3

In the final stage—Stage 3, Table 4 shows that PSO-selected features perform well across diverse classifiers (88–92.6% test accuracy, with a minor overfitting gap of 3–5%), demonstrating robust generalization. The PSO pipeline delivered notable gains for EfficientNet-B0 (+1.3%) and EfficientNetV2-B0 (+0.5%) over baselines, whereas MobileNetV3-Large showed a negligible drop, highlighting the benefit of feature optimization with different deep learning models.

Fig. 8 Confusion matrix of EfficientNetV2-B0 on test set. The image uses EfficientNetV2-B0 for main visualization; other backbone models yield similar patterns. This occurs because Blight and *Colletotrichum* share highly similar visual patterns (e.g., overlapping lesion shapes, color distributions, and texture irregularities), causing deep features to overlap in the embedding space, so without discriminative feature selection (via Particle Swarm Optimization), the model struggles to separate them effectively and produces frequent misclassifications

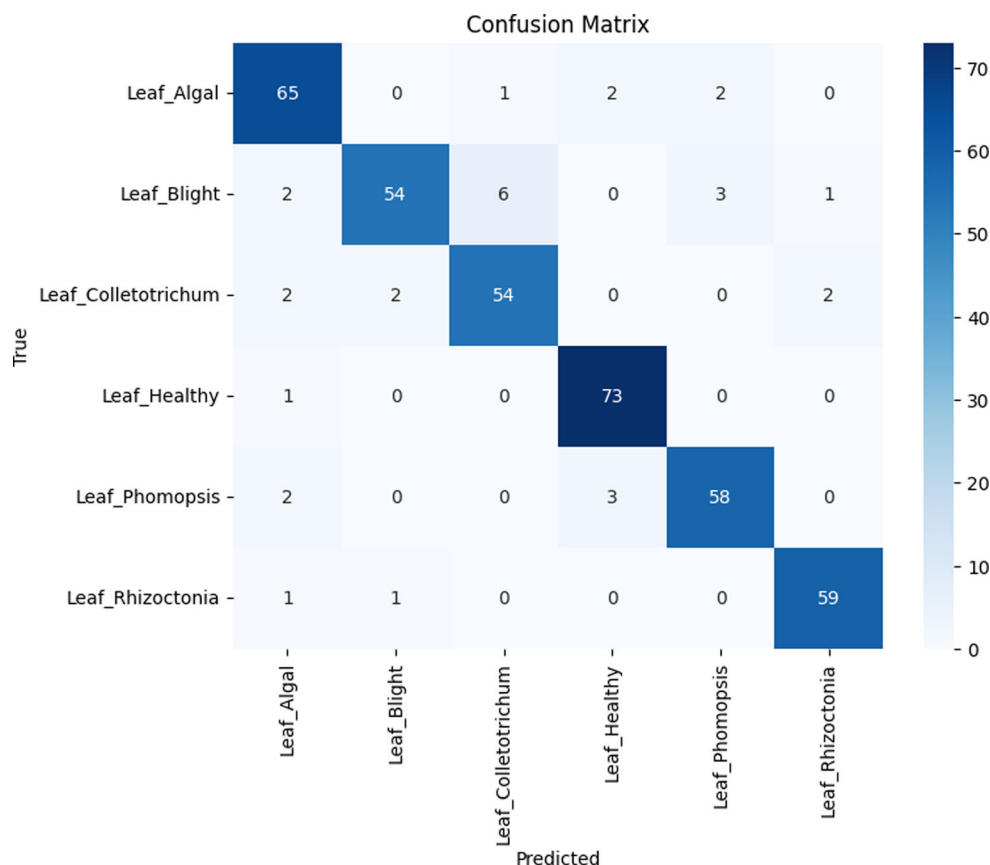


Table 4 Comprehensive classifier comparison on Particle Swarm Optimization (PSO)-selected features (phase 3). PSO improves EfficientNet models because their high-capacity feature spaces contain redundancy that can be safely pruned to enhance generalization, whereas MobileNetV3 already produces compact features, so further pruning risks removing useful information and slightly degrades performance, a trend consistently observed across classifiers

Backbone	Classifier	CV accuracy (Mean $\pm$ Std)	Test accuracy
MobileNetV3-Large	SVM-RBF	0.956 ± 0.013	0.911
	Voting-Ensemble	0.955 ± 0.013	0.906
	LogReg	0.946 ± 0.009	0.896
	Random Forest	0.944 ± 0.012	0.893
	XGBoost	0.936 ± 0.013	0.891
EfficientNet-B0	SVM-RBF	0.955 ± 0.006	0.906
	Voting-Ensemble	0.952 ± 0.004	0.896
	LogReg	0.946 ± 0.012	0.909
	Random Forest	0.944 ± 0.009	0.893
	XGBoost	0.941 ± 0.009	0.888
EfficientNetV2-B0	SVM-RBF	0.959 ± 0.006	0.924
	Voting-Ensemble	0.960 ± 0.009	0.926
	LogReg	0.957 ± 0.006	0.921
	Random Forest	0.953 ± 0.009	0.914
	XGBoost	0.950 ± 0.011	0.909

CV Cross-validation

Discussion and Limitations

This study provides the methodology for prediction and analysis using ML. Although without statistical illustrations, ML models structured in mathematics are also based on mathematical forms and included in the construction and framework in data science. However, the performance is different when using statistical concepts and production. The method should be viewed as improving efficiency and compactness, not necessarily achieving statistically significant accuracy gains. Applying PSO for feature selection on transfer learning embeddings demonstrates strong practical value by enabling more compact feature representations while maintaining robust generalization performance in the 86–92% accuracy range. Although the absolute accuracy gains are modest, the primary advantage lies in eliminating redundant features, which reduces model complexity without sacrificing predictive capability. This supports efficient deployment in lightweight environments, particularly for mobile and edge devices. From a practical perspective, the optimized feature subsets significantly reduce inference cost and computational overhead, leading to faster prediction times. In addition, the reduced feature space enhances interpretability by highlighting the most discriminative characteristics contributing to classification decisions.

Limitations:

1. The dataset focuses exclusively on leaf-based disease manifestation; diseases affecting fruits, roots, stems, or other plant organs are not covered.

2. The study is confined to durian; cross-species generalization to other crops remains unexplored.
3. The PSO pipeline achieves performance comparable to deep learning baselines; however, more advanced feature selection strategies or alternative optimization algorithms may further improve results.

Practical Validation:

Although the primary dataset used for model training and validation was derived from real-world agricultural sources, independent practical validation is necessary to assess model robustness under unseen field conditions. To ensure a fair comparison, the validation protocol follows the methodology described in Thanh et al. (2025), with minor adaptations to reflect real-world data collection constraints.

Graphical Performance

Real-world data collection (Fig. 9) was conducted during a field survey trip to Madaguoi commune, Lam Dong province, on October 20–21, 2025. The entire data collection process spanned 2 consecutive days, resulting in a diverse set of field-captured images representing authentic disease conditions in active durian cultivation areas:

- **Capture Device and Timing:** An iPhone 14 Pro Max was used to capture high-resolution leaf images. Photography sessions were conducted during early morning hours (7:00–10:00 AM) and late afternoon periods (3:00–5:00 PM) when natural lighting conditions were optimal for

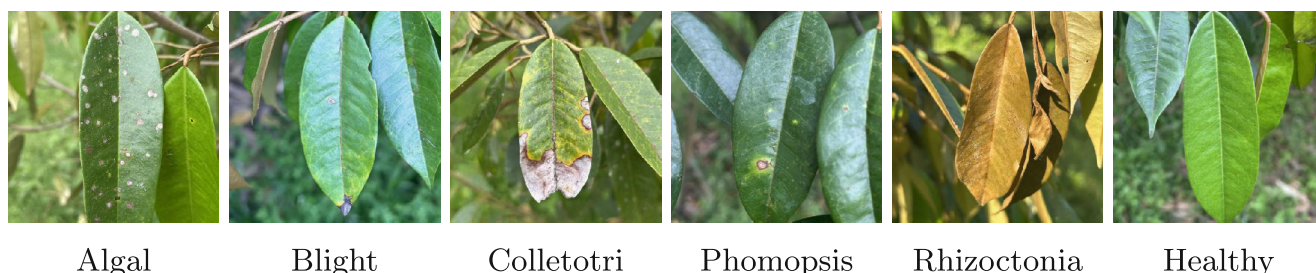
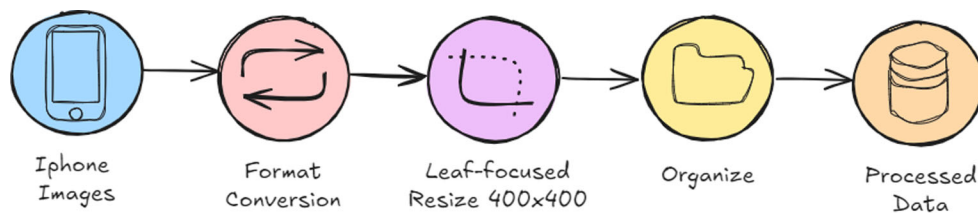
Fig. 9 Overview of the methodology for real-world dataset collection**Fig. 10** Samples from a practical dataset

image capture while avoiding harsh midday shadows and overexposure.

- **Image Capture Protocol:** Following the methodology employed by the original dataset authors, leaf samples were photographed from multiple angles and perspectives. Special attention was given to centering diseased leaf areas within the frame while ensuring adequate lighting and focus. Various background conditions were intentionally included to simulate the diverse scenarios encountered during real farm inspections.
- **Data preprocessing:** All captured images, originally in HEIC format, were converted to JPEG format, then center-cropped and resized to 400×400 pixels, ensuring compatibility with the trained model's input requirements. The preprocessed images were organized into folders based on their assigned disease labels to facilitate batch processing during model inference (98 images).
- **Data Labeling:** Images were initially classified by us based on visible symptoms. Subsequently, an agricultural expert specializing in durian cultivation and disease management in Lam Dong province carefully reviewed all labeled images and corrected any misclassifications to ensure high-quality ground truth labels for model evaluation (Fig. 10).

Practical Results

For practical validation, we validate the newly collected real-world dataset with the best models in each seed from Phase 1 only for testing. The results are summarized in Table 5, where we see that performance drops sharply on the practical set to 43–61%, highlighting overfitting or domain mismatch between the test data and real-world conditions. MobileNetV3-Large generalizes best to practical data despite not topping test metrics, while EfficientNetV2-B0 overfits the most severely, dropping over 48% in accuracy.

In detail, confusion matrices indicate near-perfect classification for LeafHealthy (19–21 TP) and LeafColletotrichum (5/6 TP), with consistently high LeafAlgal recall (17/19). LeafBlight shows the poorest recall (1–9 TP), frequently confused with Algal or *Colletotrichum* due to symptom similarities, while LeafPhomopsis varies widely strongest in MobileNetV3-Large (12 TP) and failing entirely in EfficientNetV2-B0 (0 TP). Overall, domain shift in practical data hurts blight/*Phomopsis* detection (subtle traits), prioritizing MobileNetV3 for superior generalization (Figure 11).

Table 5 Practical validation vs test set results

Model (seed)	Test set		Practical set	
	Accuracy	Macro F1	Accuracy	Macro F1
MobileNetV3-Large (512)	0.914	0.911	0.614	0.576
EfficientNet-B0 (42)	0.896	0.895	0.580	0.540
EfficientNetV2-B0 (42)	0.921	0.920	0.432	0.417

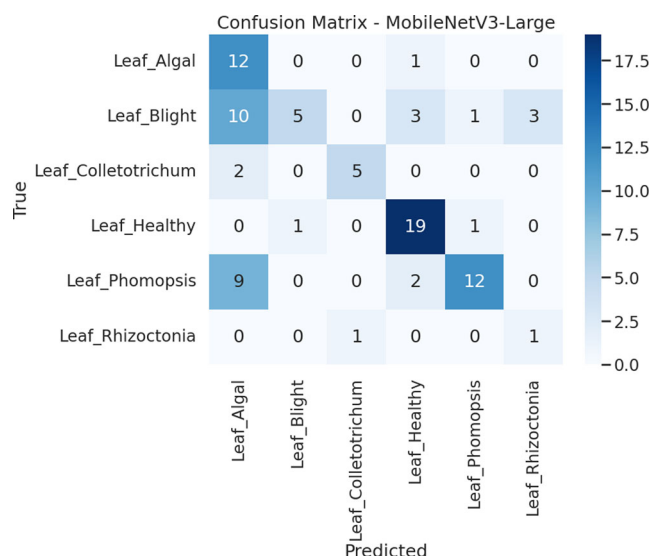


Fig. 11 Confusion matrix on the practical dataset. The image uses MobileNetV3-Large for main visualization; it showcases best performance on a practical dataset

Conclusion and Future Work

This study presented a hybrid framework combining transfer learning and Particle Swarm Optimization (PSO) for durian leaf disease classification using real-world agricultural data. The experimental results demonstrate that lightweight deep learning backbones, particularly EfficientNetV2-B0, provide strong baseline performance, while PSO-based feature selection effectively reduces feature dimensionality by approximately 48–60% with only modest numerical changes in classification accuracy. Importantly, the findings indicate that the primary contribution of PSO in this work is not substantial accuracy improvement, but rather the ability to eliminate redundant deep features and produce compact, informative representations. This reduction leads to lower computational cost, faster inference, and improved suitability for deployment on resource-constrained devices such as mobile or edge-based agricultural systems. Therefore, the proposed method should be interpreted as an efficiency-driven optimization approach rather than a statistically validated performance enhancement. Due to the absence of formal statistical significance testing, the reported improvements are best understood as numerical trends rather than conclusive gains. Furthermore, the real-world validation highlights a significant domain shift between controlled test data and field conditions, where model performance drops notably. This observation underscores a critical challenge in practical agricultural artificial intelligence applications: models trained on curated datasets may not generalize well to diverse environmental conditions, lighting variations, and complex backgrounds. Despite this limitation, MobileNetV3-Large demonstrated

relatively better robustness in real-world scenarios, suggesting that model compactness may contribute to improved generalization. Overall, the study confirms that combining deep feature extraction with metaheuristic feature selection is a viable strategy for building efficient and deployable plant disease classification systems. However, further work is required to enhance robustness, statistical validity, and real-world applicability.

Future work that will provide several directions to address the current limitations and extend the practical impact of this research includes: (1) Statistical validation and robust evaluation. Future studies should incorporate statistical significance testing (e.g., paired tests, confidence intervals, or multi-run experiments) to rigorously validate performance differences between PSO-based and baseline models. (2) Domain adaptation and generalization: To mitigate the observed domain shift, future work will explore domain adaptation techniques, including fine-tuning with field data, data normalization strategies, and distribution alignment methods to improve robustness under real-world conditions. (3) Data expansion and augmentation: Expanding the dataset to include more diverse conditions (lighting, backgrounds, growth stages) and applying data augmentation techniques will help improve model generalization and reduce overfitting. (4) Multi-organ and multi-crop extension: The current study focuses only on leaf-based diseases in durian. Future research should extend to other plant organs (fruit, stem, root) and additional crops such as coffee, rice, and pepper to evaluate cross-species applicability. (5) Advanced feature selection and optimization methods: Exploring alternative metaheuristic algorithms (e.g., genetic algorithms, differential evolution) or hybrid approaches may further improve feature selection effectiveness and model performance. (6) Real-world deployment and system optimization: Future work will include integrating the model into mobile or Internet of Things-based systems, along with evaluating system-level metrics such as inference time, memory usage, and energy efficiency to ensure practical usability for farmers. (7) Adaptive learning in field environments: Incorporating incremental learning or continuous model updating with newly collected field data can enable adaptive systems that improve over time and maintain performance across varying agricultural conditions.

Author contributions Tran Nguyen Phi Hung: Conceptualization, data curation, investigation, methodology, software, visualization, writing-original draft. Nguyen Minh Tuan: Conceptualization, data curation, investigation, methodology, software, visualization, writing-original draft and writing-review and editing, validation.

Data Availability No datasets were generated or analysed during the current study.

Competing interests The authors declare no competing interests.

References

- Adhikari S, Shrestha B, Baiju B (2018) Tomato plant diseases detection system. In: KEC Conference Proceedings, vol 1, pp 81–86
- Agarwal M, Singh A, Arjaria S, Sinha A, Gupta S (2020) ToLeD: tomato leaf disease detection using convolution neural network. *Procedia Comput Sci* 167:293–301. <https://doi.org/10.1016/j.procs.2020.03.225>
- Baser P, Saini JR, Kotecha K (2023) TomConv: an improved CNN model for diagnosis of diseases in tomato plant leaves. *Procedia Comput Sci* 218:1825–1833. <https://doi.org/10.1016/j.procs.2023.01.160>
- Bhagwat R, Dandawate Y (2021) A review of advances in automated plant disease detection. *Int J Eng Technol Innov* 11(4):251
- Brahimi M, Boukhalfa K, Moussaoui A (2017) Deep learning for tomato diseases: classification and symptoms visualization. *Appl Artif Intell* 31(4):299–315
- Das A, Pathan F, Jim JR, Kabir MM, Mridha MF (2025) Deep learning-based classification, detection, and segmentation of tomato leaf diseases: A state-of-the-art review. *Artif Intell Agric* 15(2):192–220. <https://doi.org/10.1016/j.aiia.2025.02.006>
- Elhassouny A, Smarandache F (2019) Smart mobile application to recognize tomato leaf diseases using convolutional neural networks. In: 2019 International Conference of Computer Science and Renewable Energies, pp 1–4 <https://doi.org/10.1109/ICCSRE.2019.8807737>
- Howard A, Sandler M, Chu G et al (2019) Searching for MobileNetV3. *arXiv:1905.02244*. <https://arxiv.org/abs/1905.02244>
- Howard AG, Zhu M, Chen B et al (2017) Mobilenets: efficient convolutional neural networks for mobile vision applications. *arXiv:1704.04861* (<https://arxiv.org/abs/1704.04861>)
- Kennedy J, Eberhart RC (1997) A discrete binary version of the particle swarm algorithm. In: 1997 IEEE International Conference on Systems, Man, and Cybernetics, vol 5, pp 4104–4108 <https://doi.org/10.1109/ICSMC.1997.637339>
- Khan AT, Jensen SM, Khan AR, Li S (2023) Plant disease detection model for edge computing devices. *Front Plant Sci*. <https://doi.org/10.3389/fpls.2023.1308528>
- Korf RE (1985) Depth-first iterative-deepening. *Artif Intell* 27(1):97–109. [https://doi.org/10.1016/0004-3702\(85\)90084-0](https://doi.org/10.1016/0004-3702(85)90084-0)
- Liu S, Bai H, Li F, Wang D, Zheng Y, Jiang Q, Sun F (2023) An apple leaf disease identification model for safeguarding apple food safety. *Food Sci Technol*. <https://doi.org/10.1590/fst.104322>
- Nguyen HT, Luong HH, Huynh LB, Le BQH, Doan NH, Le DTD (2023) An improved mobilenet for disease detection on tomato leaves. *Adv Technol Innov* 8(3):192–209. <https://doi.org/10.46604/aiti.2023.11568>
- Pagmo (2024) PSO visualization. <https://esa.github.io/pagmo2/docs/cpp/algorithms/psa.html>
- Paul SG, Biswas AA, Saha A, Zulfiker MS, Ritu NA, Zahan I, Rahman M, Islam MA (2023) A real-time application-based convolutional neural network approach for tomato leaf disease classification. *Array* 19:100313. <https://doi.org/10.1016/j.array.2023.100313>
- Paymode AS, Malode VB (2022) Transfer learning for multi-crop leaf disease image classification using convolutional neural network VGG. *Artif Intell Agric* 6:23–33. <https://doi.org/10.1016/j.aiia.2021.12.002>
- Pinsorn P, Sirikantaramas S (2025) From omics to orchard: The role of omics in durian cultivation. *Curr Plant Biol* 42:100466. <https://doi.org/10.1016/j.cpb.2025.100466>
- Rukuna AL, Zambuk FU, Gital AY, Bello UM (2025) Citrus diseases detection and classification based on efficientnet-B5. *Syst Soft Comput* 7:200199. <https://doi.org/10.1016/j.sasc.2025.200199>
- Sandler M, Howard A, Zhu M, Zhmoginov A, Chen L-C (2019) MobileNetV2: Inverted Residuals and Linear Bottlenecks. *arXiv:1801.04381* (<https://arxiv.org/abs/1801.04381>)
- Tan M, Le QV (2020) Efficientnet: rethinking model scaling for convolutional neural networks. *arxiv:1905.11946*. <https://arxiv.org/abs/1905.11946>
- Tan M, Le QV (2021) EfficientNetV2: Smaller Models and Faster Training. *arXiv:2104.00298*. <https://arxiv.org/abs/2104.00298>
- Thanh TN, Nguyen LX, Cap T, Le T (2025) A durian leaf image dataset of common diseases in Vietnam for agricultural diagnosis. *Data Brief* 61:111845. <https://doi.org/10.1016/j.dib.2025.111845>
- Tm P, Pranathi A, SaiAshritha K, Chittaragi NB, Koolagudi SG (2018) Tomato leaf disease detection using convolutional neural networks. In: 2018 Eleventh International Conference on Contemporary Computing (IC3), pp 1–5 <https://doi.org/10.1109/IC3.2018.8530532>
- Tuan NM, Meesad P (2025) New solutions of sixth-order Benney-Luke equation using bilinear neural network method. *Z Angew Math Phys* 76(4):133. <https://doi.org/10.1007/s00033-025-02516-8>
- Tuan NM, Meesad P, Nguyen HHC (2023) English-Vietnamese machine translation using deep learning for chatbot applications. *Sn Comput Sci* 5(1):5. <https://doi.org/10.1007/s42979-023-02339-2>
- Tuan NM, Meesad P, Hieu DV, Cuong NHH, Maliyaem M (2024) On students' sentiment prediction based on deep learning: applied information literacy. *SN Comput Sci* 5(7):928. <https://doi.org/10.1007/s42979-024-03281-7>
- Viso A (2021) Convolutional neural networks architecture. <https://viso.ai/deep-learning/vgg-very-deep-convolutional-networks/>
- Wang X, Tang SH, Ariffin MKABM, Ismail MISB, Zhao R (2025) LeafMamba: a novel IoT-integrated network for accurate and efficient plant leaf disease detection. *Alex Eng J* 123:415–424. <https://doi.org/10.1016/j.aej.2025.03.033>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.